

LECTURE 3

GEOMETRY OF LS, PROPERTIES OF $\hat{\sigma}^2$, PARTITIONED REGRESSION, GOODNESS OF FIT

Geometry of LS

We can think of y and the columns of X as members of the n -dimensional Euclidean space $\mathbb{R}^n$. One can define a subspace of $\mathbb{R}^n$ called the *column space* of an $n \times k$ matrix X , that is, the collection of all vectors in $\mathbb{R}^n$ that can be written as linear combinations of the columns of X :

$$\mathcal{S}(X) = \left\{ z \in \mathbb{R}^n : z = Xb, b = (b_1, b_2, \dots, b_k)^\top \in \mathbb{R}^k \right\}.$$

For two vectors a, b in $\mathbb{R}^n$, the distance between a and b is given by the Euclidean norm¹ of their difference $\|a - b\| = \sqrt{(a - b)^\top (a - b)}$. Thus, the LS problem, minimization of the sum of squared errors $(y - Xb)^\top (y - Xb)$, is to find, out of all elements of $\mathcal{S}(X)$, the one closest to y :

$$\min_{\tilde{y} \in \mathcal{S}(X)} \|y - \tilde{y}\|^2.$$

The closest point is found by “dropping a perpendicular”. That is, a solution to the LS problem, $\hat{y} = X\hat{\beta}$, must be chosen so that the residual vector $\hat{u} = y - \hat{y}$ is orthogonal (perpendicular) to each column of X :

$$\hat{u}^\top X = 0.$$

As a result, $\hat{u}$ is orthogonal to every element of $\mathcal{S}(X)$. Indeed, if $z \in \mathcal{S}(X)$, then there exists $b \in \mathbb{R}^k$ such that $z = Xb$, and

$$\begin{aligned} \hat{u}^\top z &= \hat{u}^\top Xb \\ &= 0. \end{aligned}$$

The collection of the elements of $\mathbb{R}^n$ orthogonal to $\mathcal{S}(X)$ is called the *orthogonal complement* of $\mathcal{S}(X)$:

$$\mathcal{S}^\perp(X) = \{z \in \mathbb{R}^n : z^\top X = 0\}.$$

Every element of $\mathcal{S}^\perp(X)$ is orthogonal to every element in $\mathcal{S}(X)$.

As we have seen in Lecture 2, the solution to the LS problem is given by

$$\begin{aligned} \hat{y} &= X\hat{\beta} \\ &= X(X^\top X)^{-1} X^\top y \\ &= P_X y, \end{aligned}$$

where

$$P_X = X(X^\top X)^{-1} X^\top$$

is called the *orthogonal projection matrix*. For any vector $y \in \mathbb{R}^n$,

$$P_X y \in \mathcal{S}(X).$$

Furthermore, the residual vector will be in $\mathcal{S}^\perp(X)$:

$$y - P_X y \in \mathcal{S}^\perp(X). \tag{1}$$

¹For a vector $x = (x_1, x_2, \dots, x_n)^\top$, its Euclidean norm is defined as $\|x\| = \sqrt{x^\top x} = \sqrt{\sum_{i=1}^n x_i^2}$.

To show (1), first note that since the columns of X are in $\mathcal{S}(X)$,

$$\begin{aligned} P_X X &= X (X^\top X)^{-1} X^\top X \\ &= X, \end{aligned}$$

and, since P_X is a symmetric matrix,

$$X^\top P_X = X^\top.$$

Now,

$$\begin{aligned} X^\top (y - P_X y) &= X^\top y - X^\top P_X y \\ &= X^\top y - X^\top y \\ &= 0. \end{aligned}$$

Thus, by definition, the residuals $y - P_X y \in \mathcal{S}^\perp(X)$. The residuals can be written as

$$\begin{aligned} \hat{u} &= y - P_X y \\ &= (I_n - P_X) y \\ &= M_X y, \end{aligned}$$

where

$$\begin{aligned} M_X &= I_n - P_X \\ &= I_n - X (X^\top X)^{-1} X^\top \end{aligned}$$

is a projection matrix onto $\mathcal{S}^\perp(X)$.

The projection matrices P_X and M_X have the following properties:

- $P_X + M_X = I$. This implies that for any $y \in \mathbb{R}^n$,

$$y = P_X y + M_X y.$$

- Symmetric:

$$\begin{aligned} P_X^\top &= P_X, \\ M_X^\top &= M_X. \end{aligned}$$

- Idempotent: $P_X P_X = P_X$, and $M_X M_X = M_X$.

$$\begin{aligned} P_X P_X &= X (X^\top X)^{-1} X^\top X (X^\top X)^{-1} X^\top \\ &= X (X^\top X)^{-1} X^\top \\ &= P_X \\ M_X M_X &= (I_n - P_X) (I_n - P_X) \\ &= I_n - 2P_X + P_X P_X \\ &= I_n - P_X \\ &= M_X. \end{aligned}$$

- Orthogonal:

$$\begin{aligned} P_X M_X &= P_X (I_n - P_X) \\ &= P_X - P_X P_X \\ &= P_X - P_X \\ &= 0. \end{aligned}$$

This property implies that $M_X X = 0$. Indeed,

$$\begin{aligned} M_X X &= (I_n - P_X) X \\ &= X - P_X X \\ &= X - X \\ &= 0. \end{aligned}$$

In the above discussion, none of the regression assumptions have been used. Given data y and X , one can always perform least squares, regardless of the data-generating process. However, a model is needed to discuss the statistical properties of an estimator (such as unbiasedness).

Properties of $\hat{\sigma}^2$

The following estimator of σ^2 was introduced in Lecture 2:

$$\begin{aligned} \hat{\sigma}^2 &= n^{-1} \sum_{i=1}^n \left(Y_i - X_i^\top \hat{\beta} \right)^2 \\ &= n^{-1} \hat{U}^\top \hat{U}. \end{aligned}$$

Under Assumptions (A1)–(A4), $\hat{\sigma}^2$ is biased. First, write

$$\begin{aligned} \hat{U} &= M_X Y \\ &= M_X (X\beta + U) \\ &= M_X U. \end{aligned}$$

The last equality follows because $M_X X = 0$. Next,

$$\begin{aligned} n\hat{\sigma}^2 &= \hat{U}^\top \hat{U} \\ &= U^\top M_X M_X U \\ &= U^\top M_X U. \end{aligned}$$

Now, since $U^\top M_X U$ is a scalar,

$$U^\top M_X U = \text{tr} (U^\top M_X U),$$

where $\text{tr}(A)$ denotes the trace of a matrix A .

$$\begin{aligned} \text{E} [U^\top M_X U \mid X] &= \text{E} [\text{tr} (U^\top M_X U) \mid X] \\ &= \text{E} [\text{tr} (M_X U U^\top) \mid X] \quad (\text{because } \text{tr}(ABC) = \text{tr}(BCA)) \\ &= \text{tr} (M_X \text{E} [U U^\top \mid X]) \quad (\text{because } \text{tr} \text{ and expectation are linear operators}) \\ &= \sigma^2 \text{tr} (M_X). \end{aligned}$$

The last equality follows because, by Assumption (A3), $\text{E} [U U^\top \mid X] = \sigma^2 I_n$. Next,

$$\begin{aligned} \text{tr} (M_X) &= \text{tr} \left(I_n - X (X^\top X)^{-1} X^\top \right) \\ &= \text{tr} (I_n) - \text{tr} \left(X (X^\top X)^{-1} X^\top \right) \\ &= \text{tr} (I_n) - \text{tr} \left((X^\top X)^{-1} X^\top X \right) \\ &= \text{tr} (I_n) - \text{tr} (I_k) \\ &= n - k. \end{aligned}$$

Thus,

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-k}{n} \sigma^2. \quad (2)$$

The estimator $\hat{\sigma}^2$ is biased, but a simple modification yields an unbiased estimator. Define

$$\begin{aligned} s^2 &= \hat{\sigma}^2 \frac{n}{n-k} \\ &= (n-k)^{-1} \sum_{i=1}^n \left(Y_i - X_i^\top \hat{\beta} \right)^2. \end{aligned}$$

It follows from (2) that

$$\mathbb{E}[s^2] = \sigma^2.$$

Partitioned regression

We can partition the matrix of regressors X as follows:

$$X = \begin{pmatrix} X_1 & X_2 \end{pmatrix},$$

and write the model as

$$Y = X_1 \beta_1 + X_2 \beta_2 + U,$$

where X_1 is an $n \times k_1$ matrix, X_2 is $n \times k_2$, $k_1 + k_2 = k$, and

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

where β_1 and β_2 are k_1 - and k_2 -vectors, respectively. Such a decomposition allows one to focus on a group of variables and their corresponding parameters, say X_1 and β_1 . If

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix},$$

then one can write the following version of the normal equations:

$$(X^\top X) \hat{\beta} = X^\top Y$$

as

$$\begin{pmatrix} X_1^\top X_1 & X_1^\top X_2 \\ X_2^\top X_1 & X_2^\top X_2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} X_1^\top Y \\ X_2^\top Y \end{pmatrix}.$$

The expressions for $\hat{\beta}_1$ and $\hat{\beta}_2$ can be obtained by inverting the partitioned matrix on the left-hand side.

Alternatively, define M_2 to be the projection matrix onto the space orthogonal to $\mathcal{S}(X_2)$:

$$M_2 = I_n - X_2 (X_2^\top X_2)^{-1} X_2^\top.$$

Then,

$$\hat{\beta}_1 = (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y. \quad (3)$$

To show this, first write

$$Y = X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2 + \hat{U}. \quad (4)$$

By construction,

$$\begin{aligned} M_2 \hat{U} &= \hat{U} \quad (\hat{U} \text{ is orthogonal to } X_2), \\ M_2 X_2 &= 0, \\ X_1^\top \hat{U} &= 0, \\ X_2^\top \hat{U} &= 0. \end{aligned}$$

Substitute equation (4) into the right-hand side of equation (3):

$$\begin{aligned} & (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 Y \\ &= (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 (X_1 \hat{\beta}_1 + X_2 \hat{\beta}_2 + \hat{U}) \\ &= (X_1^\top M_2 X_1)^{-1} X_1^\top M_2 X_1 \hat{\beta}_1 \\ &+ (X_1^\top M_2 X_1)^{-1} X_1^\top \hat{U} \quad (M_2 X_2 = 0 \text{ and } M_2 \hat{U} = \hat{U}) \\ &= \hat{\beta}_1. \end{aligned}$$

Since M_2 is symmetric and idempotent, one can write

$$\begin{aligned} \hat{\beta}_1 &= \left((M_2 X_1)^\top (M_2 X_1) \right)^{-1} (M_2 X_1)^\top (M_2 Y) \\ &= \left(\tilde{X}_1^\top \tilde{X}_1 \right)^{-1} \tilde{X}_1^\top \tilde{Y}, \end{aligned}$$

where

$$\begin{aligned} \tilde{X}_1 &= M_2 X_1 \\ &= X_1 - X_2 (X_2^\top X_2)^{-1} X_2^\top X_1 \quad (\text{residuals from the regression of } X_1 \text{ on } X_2), \\ \tilde{Y} &= M_2 Y \\ &= Y - X_2 (X_2^\top X_2)^{-1} X_2^\top Y \quad (\text{residuals from the regression of } Y \text{ on } X_2). \end{aligned}$$

Thus, to obtain the coefficients on the first k_1 regressors, instead of running the full regression with $k_1 + k_2$ regressors, one can regress Y on X_2 to obtain the residuals $\tilde{Y}$, regress X_1 on X_2 to obtain the residuals $\tilde{X}_1$, and then regress $\tilde{Y}$ on $\tilde{X}_1$ to obtain $\hat{\beta}_1$. In other words, $\hat{\beta}_1$ shows the effect of X_1 *after controlling* for X_2 .

Similarly to $\hat{\beta}_1$, one can write:

$$\begin{aligned} \hat{\beta}_2 &= (X_2^\top M_1 X_2)^{-1} X_2^\top M_1 Y, \text{ where} \\ M_1 &= I_n - X_1 (X_1^\top X_1)^{-1} X_1^\top. \end{aligned}$$

For example, consider a simple regression

$$Y_i = \beta_1 + \beta_2 X_i + U_i,$$

for $i = 1, \dots, n$.

Define an n -vector of ones:

$$\ell = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

In this case, the matrix of regressors is given by

$$\begin{pmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{pmatrix} = \begin{pmatrix} \ell & X \end{pmatrix}.$$

Consider

$$M_1 = I_n - \ell (\ell^\top \ell)^{-1} \ell^\top,$$

and

$$\hat{\beta}_2 = \frac{X^\top M_1 Y}{X^\top M_1 X}.$$

Now, $\ell^\top \ell = n$. Therefore,

$$\begin{aligned} M_1 &= I_n - \frac{1}{n} \ell \ell^\top, \text{ and} \\ M_1 X &= X - \ell \frac{\ell^\top X}{n} \\ &= X - \bar{X} \ell \\ &= \begin{pmatrix} X_1 - \bar{X} \\ X_2 - \bar{X} \\ \vdots \\ X_n - \bar{X} \end{pmatrix}, \end{aligned}$$

where

$$\begin{aligned} \bar{X} &= \frac{\ell^\top X}{n} \\ &= n^{-1} \sum_{i=1}^n X_i. \end{aligned}$$

Thus, the matrix M_1 transforms the vector X into the vector of deviations from the average. We can write

$$\begin{aligned} \hat{\beta}_2 &= \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ &= \frac{\sum_{i=1}^n (X_i - \bar{X}) (Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}. \end{aligned}$$

Goodness of fit

Write

$$\begin{aligned} Y &= P_X Y + M_X Y \\ &= \hat{Y} + \hat{U}, \end{aligned}$$

where, by construction,

$$\begin{aligned} \hat{Y}^\top \hat{U} &= (P_X Y)^\top (M_X Y) \\ &= Y^\top P_X M_X Y \\ &= 0. \end{aligned}$$

Suppose that the model contains an intercept, i.e., the first column of X is the vector of ones ℓ . The *total variation* in Y is

$$\begin{aligned}\sum_{i=1}^n (Y_i - \bar{Y})^2 &= Y^\top M_1 Y \\ &= (\hat{Y} + \hat{U})^\top M_1 (\hat{Y} + \hat{U}) \\ &= \hat{Y}^\top M_1 \hat{Y} + \hat{U}^\top M_1 \hat{U} + 2\hat{Y}^\top M_1 \hat{U}.\end{aligned}$$

Since the model contains an intercept,

$$\begin{aligned}\ell^\top \hat{U} &= 0, \text{ and} \\ M_1 \hat{U} &= \hat{U}.\end{aligned}$$

However, $\hat{Y}^\top \hat{U} = 0$, and, therefore,

$$\begin{aligned}Y^\top M_1 Y &= \hat{Y}^\top M_1 \hat{Y} + \hat{U}^\top \hat{U}, \text{ or} \\ \sum_{i=1}^n (Y_i - \bar{Y})^2 &= \sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2 + \sum_{i=1}^n \hat{U}_i^2.\end{aligned}$$

Note that

$$\begin{aligned}\bar{Y} &= \frac{\ell^\top Y}{n} \\ &= \frac{\ell^\top \hat{Y}}{n} + \frac{\ell^\top \hat{U}}{n} \\ &= \frac{\ell^\top \hat{Y}}{n} \\ &= \bar{\hat{Y}}.\end{aligned}$$

Hence, the averages of Y and its predicted values $\hat{Y}$ are equal, and we can write:

$$\begin{aligned}\sum_{i=1}^n (Y_i - \bar{Y})^2 &= \sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2 + \sum_{i=1}^n \hat{U}_i^2, \text{ or} \\ TSS &= ESS + RSS,\end{aligned}\tag{5}$$

where

$$\begin{aligned}TSS &= \sum_{i=1}^n (Y_i - \bar{Y})^2 \text{ (total sum of squares),} \\ ESS &= \sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2 \text{ (explained sum of squares),} \\ RSS &= \sum_{i=1}^n \hat{U}_i^2 \text{ (residual sum of squares).}\end{aligned}$$

The ratio of ESS to TSS is called the *coefficient of determination*, or R^2 :

$$\begin{aligned} R^2 &= \frac{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \\ &= 1 - \frac{\sum_{i=1}^n \hat{U}_i^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2} \\ &= 1 - \frac{\hat{U}^\top \hat{U}}{Y^\top M_1 Y}. \end{aligned}$$

Properties of R^2 :

- R^2 is bounded between 0 and 1, as implied by decomposition (5). This property does not hold if the model lacks an intercept; one should not use the above definition of R^2 in that case. If $R^2 = 1$, then $\hat{U}^\top \hat{U} = 0$, which can happen only if $Y \in \mathcal{S}(X)$, i.e., Y is *exactly* a linear combination of the columns of X .
- R^2 increases when additional regressors are added.

Proof. Consider a partitioned matrix $X = \begin{pmatrix} Z & W \end{pmatrix}$. We study the effect of adding W on R^2 . Let

$$\begin{aligned} P_X &= X (X^\top X)^{-1} X^\top \text{ (projection matrix for the full regression),} \\ P_Z &= Z (Z^\top Z)^{-1} Z^\top \text{ (projection matrix for the regression without } W\text{).} \end{aligned}$$

Define also

$$\begin{aligned} M_X &= I_n - P_X, \\ M_Z &= I_n - P_Z. \end{aligned}$$

Since Z is part of X ,

$$P_X Z = Z,$$

and

$$\begin{aligned} P_X P_Z &= P_X Z (Z^\top Z)^{-1} Z^\top \\ &= Z (Z^\top Z)^{-1} Z^\top \\ &= P_Z. \end{aligned}$$

Consequently,

$$\begin{aligned} M_X M_Z &= (I_n - P_X) (I_n - P_Z) \\ &= I_n - P_X - P_Z + P_X P_Z \\ &= I_n - P_X - P_Z + P_Z \\ &= M_X. \end{aligned}$$

Assume that Z contains a column of ones, so both short and long regressions have intercepts. Define

$$\begin{aligned} \hat{U}_X &= M_X Y, \\ \hat{U}_Z &= M_Z Y. \end{aligned}$$

Write:

$$\begin{aligned} 0 &\leq (\hat{U}_X - \hat{U}_Z)^\top (\hat{U}_X - \hat{U}_Z) \\ &= \hat{U}_X^\top \hat{U}_X + \hat{U}_Z^\top \hat{U}_Z - 2\hat{U}_X^\top \hat{U}_Z. \end{aligned}$$

Next,

$$\begin{aligned}\hat{U}_X^\top \hat{U}_Z &= Y^\top M_X M_Z Y \\ &= Y^\top M_X Y \\ &= \hat{U}_X^\top \hat{U}_X.\end{aligned}$$

Hence,

$$\hat{U}_Z^\top \hat{U}_Z \geq \hat{U}_X^\top \hat{U}_X.$$

- R^2 shows how much of the *sample* variation in y was explained by X . However, our objective is to estimate *population* relationships, not to explain *sample* variation. A high R^2 does not necessarily indicate a good regression model, and a low R^2 is not evidence against one.
- One can always find an X that makes $R^2 = 1$: take any n linearly independent vectors. Since such a set spans $\mathbb{R}^n$, any $y \in \mathbb{R}^n$ can be written as an exact linear combination of the columns of X .

Since R^2 increases with the inclusion of additional regressors, researchers often report instead the *adjusted coefficient of determination* $\bar{R}^2$:

$$\begin{aligned}\bar{R}^2 &= 1 - \frac{n-1}{n-k} (1 - R^2) \\ &= 1 - \frac{\hat{U}^\top \hat{U} / (n-k)}{Y^\top M_1 Y / (n-1)}.\end{aligned}$$

The adjusted coefficient of determination penalizes the fit when the number of regressors k is large relative to the number of observations n , and $\bar{R}^2$ may decrease as k increases. However, there is no strong argument for using this adjustment.